LLM Cost Estimate for Literary Analysis Project

Per-Poem Token Estimates

Input: Average Poem

- **Average word count:** 200 words (liberal estimate)
- **Average characters per word:** 5
- **Total characters:** 200 × 6 = 1,200 characters
- **Tokens (measured):** ~201 tokens

Prompt 0: Prior Knowledge Check

- Instruction text: ~39 tokens
- Schema (`CheckPriorKnowledgeInference`): ~77 tokens
- Poem text: ~201 tokens
- **Total input:** ~317 tokens per poem

Prompt 1: Literary Devices Inference

- Instruction text + devices list (35 devices): ~149 tokens
- Schema (`LiteraryDevicesInference`): ~139 tokens
- Poem text: ~201 tokens
- **Total input:** ~489 tokens per poem

Prompt 2: Score Inference

- Instruction & rubric: ~78 tokens
- Schema (`ScoreInference`): ~293 tokens
- Poem text: ~201 tokens
- **Total input:** ~572 tokens per poem

Expected Outputs (per inference)

- **Prior Knowledge Check:** ~10 tokens
- **Literary Devices:** Rationale (~200 tokens) + device list (~20 tokens) = ~220 tokens
- **Score:** Rationale (~300 tokens) + 5 numerical scores (~10 tokens) = ~310 tokens

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Cost Breakdown per Poem (2 inferences × 3 models = 6 API calls)

Input Tokens (all models)

- Prior Knowledge Check: 317 tokens × 3 models = 951 tokens
- Literary Devices: 489 tokens × 3 models = 1,467 tokens
- Scoring: 572 tokens × 3 models = 1,716 tokens
- **Total input: 4,134 tokens per poem**

Output Tokens (estimates)

- Prior Knowledge Check: 10 tokens × 3 models = 30 tokens
- Literary Devices: 220 tokens × 3 models = 660 tokens
- Scoring: 310 tokens × 3 models = 930 tokens
- **Total output: 1,620 tokens per poem**

Pricing (estimated 2026 rates)

Model Pricing (per 1M tokens)

- **GPT-5.4:** Input $2.50, Output $15.00
- **Claude 4.6 Opus:** Input $5.00, Output $25.00
- **Gemini 3.1 Pro:** Input $2.00, Output $12.00

Cost per Poem (per model)

Per-model input: 1,378 tokens. Per-model output: 540 tokens.

Model	Rate (Input / Output per 1M)	Input Cost	Output Cost	Total Cost
GPT-5.4	$2.50 / $15.00	$0.003445	$0.008100	**$0.011545**
Claude Opus	$5.00 / $25.00	$0.006890	$0.013500	**$0.020390**
Gemini Pro	$2.00 / $12.00	$0.002756	$0.006480	**$0.009236**

Cost per Poem (all 3 models): ~$0.041171

Scaling to Full Dataset

Poem Count	Total Cost	GPT-5.4	Claude	Gemini	Notes
10	$0.41	$0.12	$0.20	$0.09	Test batch
50	$2.06	$0.58	$1.02	$0.46	
100	$4.12	$1.15	$2.04	$0.92	
500	$20.59	$5.77	$10.20	$4.62	
1,000	$41.17	$11.55	$20.39	$9.24	Full-scale run

Full-Scale Allocation With Leeway (1,000 Poems)

Baseline full-scale API cost is **$41.17**.

Budget by Leeway Level

Leeway	Formula	Total Budget

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0%	$41.17 × 1.00	**$41.17**
15%	$41.17 × 1.15	**$47.35**
25%	$41.17 × 1.25	**$51.46**
40%	$41.17 × 1.40	**$57.64**

Specific Allocation (Recommended: 25% Leeway)

Using a **$51.46** budget cap:

- **Core planned inference (all poems, all models):** $41.17
- **Retry / transient API failures reserve (10%):** $4.12
- **Longer-than-expected outputs reserve (7.5%):** $3.09
- **Prompt iteration / QA spot-check reserve (7.5%):** $3.09

Total allocated: **$51.47**

Model-Level Allocation at 25% Leeway

Model	Baseline (1,000 poems)	With 25% Leeway
GPT-5.4	$11.55	$14.43
Claude 4.6 Opus	$20.39	$25.49
Gemini 3.1 Pro	$9.24	$11.55
Total	**$41.17**	**$51.46**

Note: Current `data/poems.csv` has 1 poem row, so this is a projection for the full target dataset.